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An organic small molecule fluorescent probe for nondestructive detection of Zn^{2+} in plants

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ABSTRACT

In this study, a novel two-photon fluorescent probe, **ZT1**, leveraging the photoinduced electron transfer (PET) effect was designed and synthesized. The probe employs 4-trifluoromethyl coumarin as the fluorophore and 2,2′-dipicolylamine (DPA) as the Zn^{2+} recognition group, enabling non-destructive detection of Zn^{2+} in *Arabidopsis thaliana*. **ZT1** demonstrated exceptional selectivity for Zn^{2+} in both *in vitro* and *in vivo* assays, maintained stability across a pH range of 6–8, featured a detection limit as low as 13.96 nM, and exhibited a large Stokes shift (177 nm). The probe facilitated high-resolution visualization of Zn^{2+} in the root tips and leaf tissues of *Arabidopsis thaliana* and demonstrated remarkable reversibility in cyclic experiments. This work establishes a robust toolkit that overcomes longstanding limitations in plant metal ion imaging, offering mechanistic insights into zinc-regulated physiological processes.

Introduction

Zinc ions (Zn^{2+}) are indispensable micronutrients for plant growth and development, exerting significant influence on a myriad of physiological and biochemical processes such as enzyme activation, protein synthesis, hormone regulation, and responses to oxidative stress [1]. However, aberrant Zn^{2+} concentrations can detrimentally impact plant growth. Deficient Zn^{2+} levels can inhibit auxin synthesis and disrupt phosphate metabolism, thereby impeding plant development. On the other hand, excessive Zn^{2+} concentrations can impede the uptake of other essential ions and induce oxidative stress through the overproduction of reactive oxygen species (ROS), resulting in oxidative damage and potentially plant toxicity [2,3]. The precise quantification and mapping of Zn^{2+} levels within plants are paramount for elucidating plant physiological states and zinc metabolism. These insights are instrumental not only for optimizing agricultural productivity but also for environmental conservation and plant science research [4,5]. Consequently, the development of real-time, precise techniques for detecting Zn^{2+} in plants presents a critical challenge that warrants immediate attention.

A variety of methodologies have been effectively utilized for the detection of Zn^{2+} levels in plants, encompassing atomic absorption spectroscopy (AAS) [6], atomic emission spectroscopy (AES) [7], inductively coupled plasma mass spectrometry (ICP-MS) [8], and electrochemical techniques [9]. Despite the notable potency of these methodologies, it is evident that certain limitations persist, such as prohibitive costs, operational intricacy, and an inability to facilitate visualization or non-destructive detection. Thus, fluorescent probe technology has garnered significant attention due to its capacity for real-time, non-invasive, highly sensitive, and selective detection [10–13],

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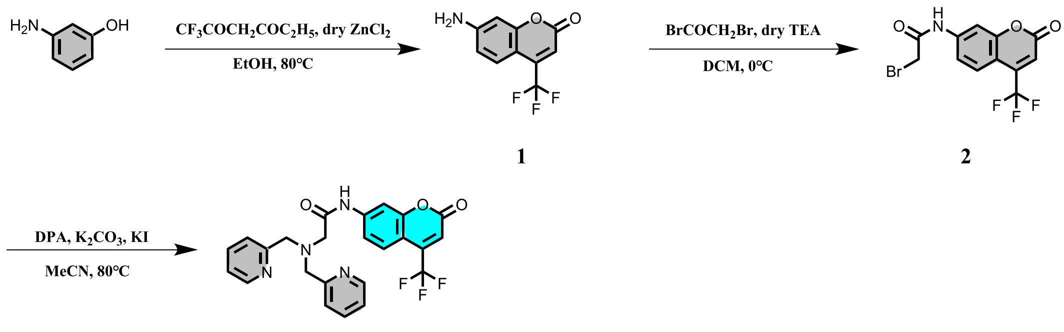

Scheme 1. The synthetic route of probe **ZT1**.

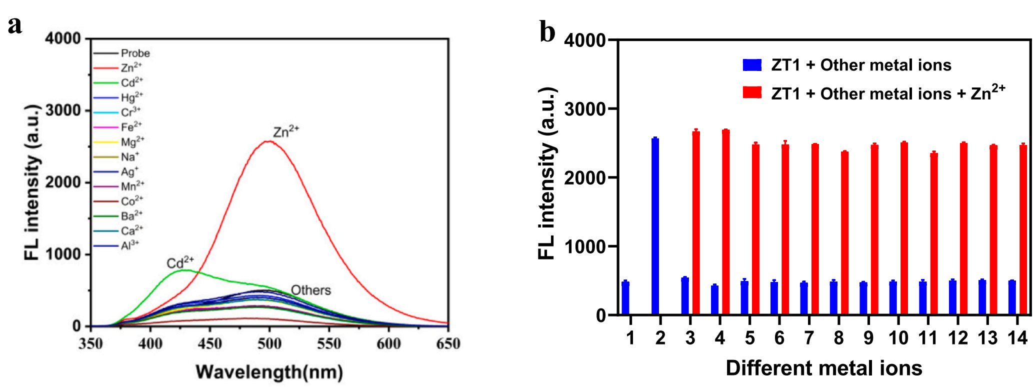

Fig. 1. a) Fluorescent response of **ZT1** to various metal ions; b) Fluorescent selectivity of **ZT1** with Zn^{2+} in the presence of various metal ions, including: 1-**ZT1**, 2-Zn^{2+}, 3-Cd^{2+}, 4-Hg^{2+}, 5-Cr^{3+}, 6-Fe^{2+}, 7-Mg^{2+}, 8-Na^{+}, 9-Ag^{+}, 10-Mn^{2+}, 11-Co^{2+}, 12-Ba^{2+}, 13-Ca^{2+}, 14-Al^{3+}.

probes based on different fluorophores are constantly reported in recent years [14–18]. Among the spectrum of fluorescent probes, two-photon fluorescence technology stands out for its use of near-infrared (NIR) light as the excitation source, which confers substantial benefits, including deeper tissue penetration, diminished photo-damage, and superior resolution [19–25]. Additionally, it effectively mitigates interference from plant's auto-fluorescence. Recent years have seen the emergence of two-photon fluorescent probes, which demonstrated remarkable efficacy in the detection of heavy metal ions, such as Cu^{2+}, in plant specimens, facilitating high-resolution visual imaging [26]. However, research on two-photon fluorescent probes for the detection of Zn^{2+} levels and distribution in plants remains scarce [27]. Consequently, the development of an efficient two-photon fluorescent probe for Zn^{2+} detection in plants presents a significant opportunity to advance the frontiers of plant science research.

In recent years, there has been considerable progress in the development of a wide variety of two-photon fluorescent probes, which are predicated on a number of fluorophores including 1,8-naphthalimide [28,29], anthracene [30], coumarin [31], benzothiazole [32] and rhodamine [33], with applications in biological imaging. Among these, coumarin derivatives, particularly those with an electron-withdrawing group at the 4-position and an amine group at the 7-position, have garnered considerable interest due to their substantial Stokes shift, high quantum yield, excellent biocompatibility, cell membrane permeability, and ease of modification [34–37]. In this study, we opted for a coumarin derivative featuring a trifluoromethyl group at the 4-position and an amine group at the 7-position as the fluorophore. 2,2′-dipicolylamine (DPA) was selected as the recognition group for its high selectivity toward Zn^{2+}. The probe, ZT1, was engineered based on the photoinduced electron transfer (PET) effect. In addition to its superior water solubility, the probe also exhibited remarkable selectivity for Zn^{2+} in both *in vitro* and *in vivo* assays. Furthermore, **ZT1** demonstrated a favourable linear response within the 0–0.5 μM range and a low detection limit of 13.96 nM. Notably, **ZT1** showed exceptional performance in high-resolution subcellular imaging of Arabidopsis root and leaf tissues, successfully delineated the spatiotemporal distribution of Zn^{2+} in these tissues. Collectively, **ZT1** offered a potent tool for elucidating plant physiological states and unraveling zinc metabolism mechanisms. Additionally, the probe displayed promise in the domains of plant science, food production, and agricultural planting, with the potential to enhance cost efficiency and productivity in these sectors.

Results and discussion

Spectral characteristics of probe ZT1 toward Zn^{2+}

The synthetic route of probe **ZT1** is shown in Scheme 1. The synthetic procedure of target molecule is outlined as follow: 2-bromo-*N*-(2-oxo-4-(trifluoromethyl)-2H-chromen-7-yl) acetamide (0.20 g, 0.57

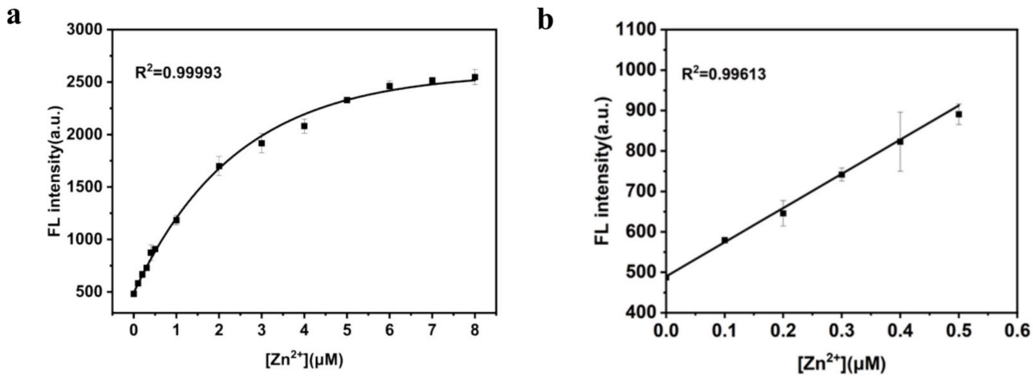

Fig. 2. a) Fluorescent response of the probe (1 μ M) to Zn^{2+} (0–8 μ M); b) Linear relationship between fluorescence intensity and Zn^{2+} concentration (0–0.5 μ M).

mmol), 2, 2′-dipicolylamine (0.11 g, 0.57 mmol), K_2CO_3 (0.16 g, 1.14 mmol) and a small amount of KI were dissolved in anhydrous 5 mL CH_3CN. The mixture was refluxed at 60 °C for 6 h under the protection of N_2. The crude product was purified on a silica gel column using dichloromethane/methanol (50,1) as eluent. After removing the solvent under reduced pressure, the final product **ZT1** was obtained as white solid (0.034 g, 12.7 %). ^{1}H NMR (600 MHz, Chloroform-*d*) δ 11.57 (s, 1H), 8.67–8.63 (m, 2H), 7.97 (d, J = 2.0 Hz, 1H), 7.85 (dd, J = 8.8, 2.0 Hz, 1H), 7.68 (dd, J = 8.8, 1.5 Hz, 1H), 7.63 (ddd, J = 7.6, 1.8 Hz, 2H), 7.25 (d, J = 7.7 Hz, 2H), 7.23–7.20 (m, 2H), 6.67 (s, 1H), 3.94 (s, 4H), 3.53 (s, 2H). ^{13}C NMR (100 MHz, Chloroform-*d*) δ 169.86, 158.42, 156.81, 154.32, 148.44, 142.31, 135.80, 124.82, 124.80, 122.34, 121.75, 115.46, 112.43, 112.37, 108.10, 106.36, 59.26, 57.88. ^{19}F NMR (565 MHz, DMSO-d_6) δ −63.57. Calculated for **ZT1** + $\mathbf{H^+}$: 469.1483, found: 469.1477. Other synthesis procedure and the characterization results of compounds are shown in the Supporting Information (Fig. S1–10).

In HEPES buffer solution (10 mM, pH 7.4), upon irradiation with a 333 nm laser (Fig. S11), the probe **ZT1** displayed a pronounced fluorescence absorption peak at approximately 510 nm. To ascertain the selectivity of **ZT1** for various metal ions, its fluorescence response to a panel of ions including K^+, Cd^{2+}, Hg^{2+}, Zn^{2+}, Na^+, Ag^+, Cr^{3+}, Fe^{2+}, Mg^{2+}, Mn^{2+}, Co^{2+}, Ba^{2+}, Ca^{2+}, and Al^{3+} was evaluated, as depicted in Fig. 1a. The experimental findings indicate that **ZT1** exhibited negligible reactivity to other metal ions, while its response to Zn^{2+} is markedly robust. Additionally, the probe shows a modest response to Cd^{2+}, accompanied by a blue shift in the emission spectrum, aligning with prior literature. These outcomes, therefore, substantiated the exceptional selectivity of **ZT1** for Zn^{2+}. In natural environments, plants are frequently exposed to a cocktail of heavy metal stresses rather than a solitary metal stress. To probe the selectivity of **ZT1** under conditions of concurrent metal ion presence, metal ion coexistence titration experiments were performed. The experimental data (Fig. 1b) revealed that even in the presence of other metal ions, **ZT1** maintained its excellent selectivity for Zn^{2+}. This finding further corroborated the superior selectivity for Zn^{2+}.

To elucidate the response of probe **ZT1** to varying concentrations of Zn^{2+}, we monitored the fluorescence intensity changes at 510 nm upon interaction with Zn^{2+}. As illustrated in Fig. 2a, within the concentration range of 0 to 8 μM Zn^{2+}, the fluorescence intensity escalated progressively with increasing Zn^{2+} concentration. By analyzing the experimental data and conducting nonlinear regression, it was observed that the fluorescence intensity of **ZT1** increased by over fivefold in response to Zn^{2+}. Initially, the fluorescence intensity surged rapidly, then plateaued, and approached saturation. Subsequently, **ZT1** was incubated with a series of Zn^{2+} concentrations, and the fluorescence emission intensity at 510 nm was measured and correlated with Zn^{2+} concentration. As depicted in Fig. 2b, for Zn^{2+} concentrations ranging from 0 to 0.5 μM, the fluorescence intensity of the solution system demonstrated a robust linear relationship with Zn^{2+} concentration. The linear regression equation was: $y = 845.3990\times + 489.9363$. Utilizing the formula LOD = K Sb/M, where K is a constant, Sb is the standard deviation of the blank, and M is the slope of the calibration curve, the limit of detection (LOD) for **ZT1** toward Zn^{2+} was calculated to be LOD = 3 × 3.9330 / 845.3990 = 0.01396 μM. That is 13.96 nM, which conclusively demonstrates **ZT1**'s exceptional sensitivity toward Zn^{2+}. Comparative analysis with contemporary zinc-responsive probes further confirms **ZT1**'s superior aqueous solubility, obviating the need for organic solvent coadministration during analytical procedures. This unique characteristic significantly enhances **ZT1**'s viability for non-destructive *in situ* monitoring of Zn^{2+} spatial distribution in plant tissues (Table S8).

Mechanism of recognition of ZT1 to Zn^{2+}

To elucidate the response mechanism of probe **ZT1** to Zn^{2+}, a hypothesis was posited as depicted in Fig. 3a. To delve deeper into the interaction between **ZT1** and Zn^{2+}, theoretical calculations of the Frontier Molecular Orbitals (FMO) for both **ZT1** and its zinc complex (**ZT1**-Zn^{2+}) were conducted at the B3LYP/def2svp computational level. As shown in Fig. 3b, the HOMO-LUMO gap expanded upon complexation with the zinc ion, which depends on the attraction of Zn^{2+} to **ZT1** electrons according to the Mulliken population analysis and Natural Bond Orbital analysis (refer to Fig. S16 and Table S3-S7). And the FMO energy levels of **ZT1**-Zn^{2+} are lower than those of **ZT1** for stabilization. Moreover, the electron density of the HOMO-LUMO of **ZT1** was distributed across the PDA and coumarin moieties, respectively. While in the **ZT1**-Zn^{2+} complex, both FMOs are localized within the PDA moiety. Consequently, the mechanism of **ZT1** aligns with the photoinduced electron transfer (PET) effect.

As shown in Figure S11, the **ZT1**-Zn^{2+} complex exhibits a 4 nm blue shift in its ultraviolet absorption maximum ($\Delta\lambda$ = 337 nm → 333 nm), while its fluorescence emission peak remains stable at 501 ± 2 nm (Figure S16). This behavior aligns with the spectral trends previously reported for photoinduced electron transfer (PET)-based probes [38,39], yet demonstrates a significantly weaker displacement magnitude compared to the typical wavelength shifts observed in chelation-enhanced fluorescence (CHEF) probe systems [40,41].

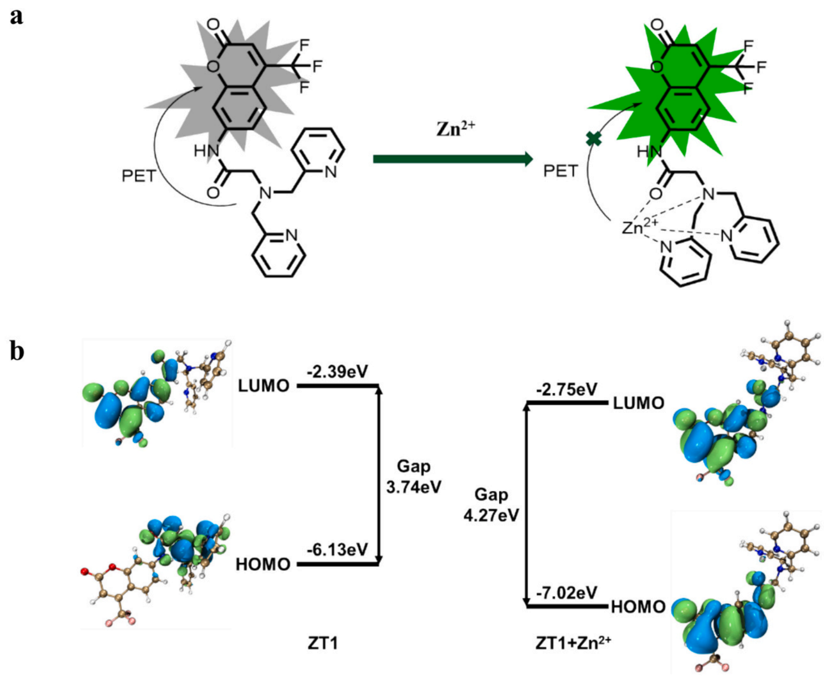

Fig. 3. a) The possible mechanism of the **ZT1**; b) HOMO and LUMO energy levels and the molecular orbital plots of **ZT1** and **ZT1**-Zn^{2+}. The structures of **ZT1** and **ZT1**-Zn^{2+} are obtained by conformational search using xtb 6.7.1 [42], Molclus 1.12 [43], ORCA 6.0.0 [44], and the plot was drawn using Multiwfn [45,46] and VMD 1.9.3 [47].

The recognition mechanism of the probe was further investigated using nuclear magnetic resonance (NMR) hydrogen spectra (refer to Fig. S12) before and after **ZT1** binding to Zn^{2+}. The experimental data revealed shifts in the chemical shifts of the hydrogen atoms on the pyridine ring and amide, confirmed the complexation of DPA with Zn^{2+}. As illustrated in Fig. S13, the binding stoichiometry between **ZT1** and Zn^{2+} was determined to be 1:1 using the Job's plot method. This stoichiometry was further corroborated by high-resolution mass spectrometry (HRMS) spectra (refer to Fig. S14).

In Vivo Imaging of Zn^{2+}

The limited penetration depth of laser scanning confocal microscopy (LSCM) restricts its ability to probe the distribution of Zn^{2+} within plant cells. In contrast, two-photon confocal laser microscopy (TPCLM), which employs dual near-infrared excitation sources, offers distinct advantages. TPCLM not only enhances spatial localization in marker detection but also simplifies the optical design. Additionally, it generates stronger fluorescence signals, leading to a higher signal-to-noise ratio (SNR) in the resulting images.

Arabidopsis thaliana plants, cultivated in an artificial climate chamber, were meticulously extracted from their growth medium using forceps and subsequently immersed individually in solutions containing Na^{+}, Mg^{2+}, Cd^{2+}, Hg^{2+}, and Zn^{2+} (each at 100 μM) for a 2-h incubation period. The root tips and leaf tissues were then rinsed with distilled water and gently blotted dry with absorbent paper to remove excess moisture. The plants were subsequently immersed in a pre-prepared **ZT1** solution (10 μM) for 1 h, followed by another rinse with distilled water. The Arabidopsis plants were finally mounted on glass slides, covered with cover slips, and imaged using a microscope to examine the root tips and leaf tissues.

To further elucidate the fluorescence signal transmission process in Arabidopsis under varying Zn^{2+} concentrations, TPCLM imaging was conducted on Arabidopsis root tips and leaf tissues treated with a range of Zn^{2+} concentrations.

Initially, Arabidopsis plants were subjected to Zn^{2+} solutions of varying concentrations for a 2-h period. Following this exposure, the tissues designated for imaging were rinsed with distilled water and then incubated in a prepared probe solution for 1 h. Post-treatment, the plants were mounted on glass slides, covered with cover slips, and prepared for imaging.

As depicted in Fig. 4a, the control group, devoid of Zn^{2+}, exhibited faint fluorescence signals in the fluorescence imaging. In Arabidopsis plants treated with 10 μM Zn^{2+}, **ZT1** commenced to exhibit fluorescence signals circumscribing the root tip and leaf cells. Upon treatment with 100 μM Zn^{2+}, the fluorescence signals in both root tips and leaves were markedly intensified. Ultimately, after treatment with 1000 μM Zn^{2+}, the fluorescence intensity in both root tips and leaves peaked. According to the statistical data (Fig. 4b and c), the similar conclusion can be drawn.

These findings indicated that with escalating Zn^{2+} concentrations, there was a proportional increase in the binding affinity of **ZT1** to Zn^{2+} and the corresponding *in vivo* fluorescence intensity. **ZT1** thus manifested exceptional sensitivity in visualizing Zn^{2+}-induced stress in plants.

In the control group, where no Zn^{2+} was employed for incubation, the fluorescence images (Fig. 5a) revealed only faint fluorescence signals. Following a 1-h incubation of Arabidopsis root tips in the Zn^{2+} solution, imaging results indicated that fluorescence signals, indicative of the probe's interaction with Zn^{2+}, were detectable around the epidermal cells of the root tip. After a 3-h incubation in the Zn^{2+} solution, pronounced fluorescence signals stemming from the probe-Zn^{2+} complex were observed across most areas of the root tip cells. By the 5-h mark, the fluorescence intensity within the root tip cells had substantially increased, exhibiting an excellent signal-to-noise ratio. Analogous observations were made in TPCLM imaging of Arabidopsis leaves. According to the statistical data (Fig. 5b and c), the similar conclusion can be drawn.

These findings underscored the **ZT1** probe's capability to detect the distribution of Zn^{2+} in Arabidopsis root tips and leaves in real time. Across varying incubation durations, the **ZT1** probe and Zn^{2+} progressively infiltrated the root tip and leaf cells. In the root tip, Zn^{2+} initially accumulated in the outer epidermal cells and, with time, gradually penetrated into the root tip cells, eventually diffusing throughout the

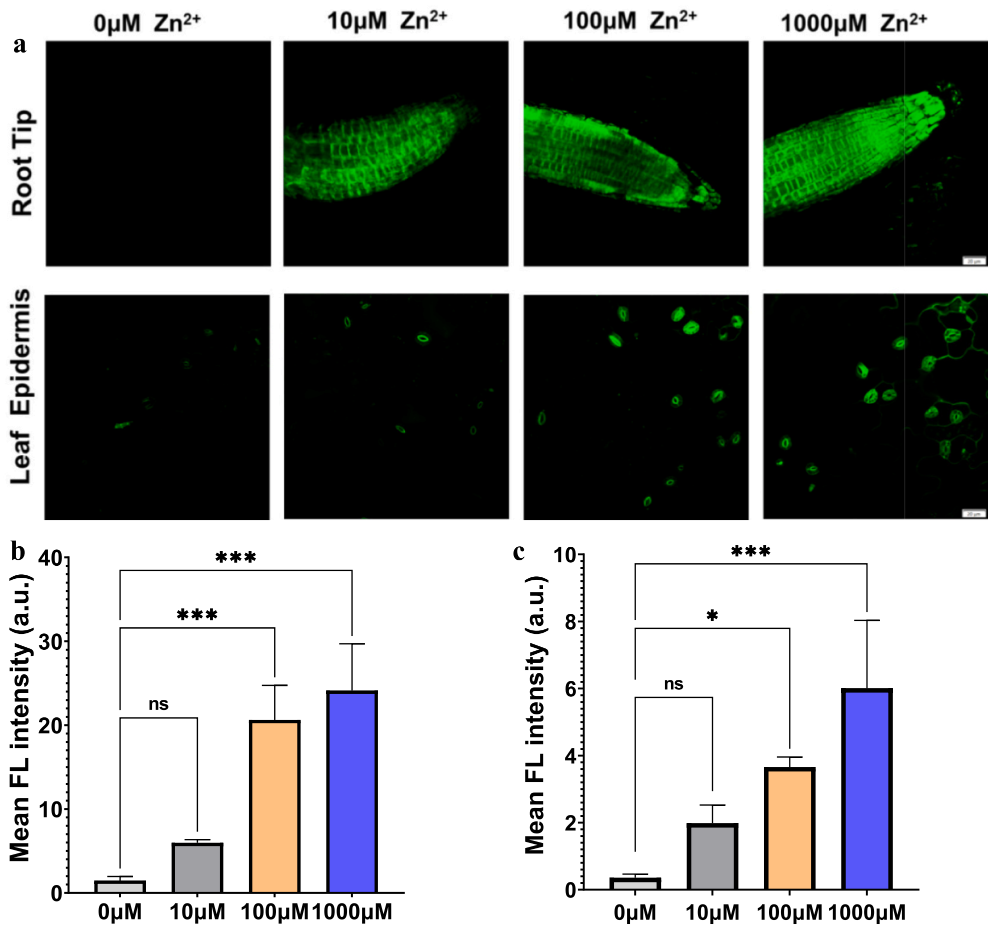

Fig. 4. a) Two-photon imaging of the probe (10 μM) in Arabidopsis root tips and leaves (Simple numbers $n = 3$, scale bar = 20 μm) at different concentrations; b) Average fluorescence intensity of Arabidopsis root tips; c) Average fluorescence intensity of Arabidopsis leaves.

root tip. In the leaves, Zn^{2+} initially accumulated around the stomata, and as time elapsed, it gradually permeated into the leaf cells, culminating in a uniform distribution across the entire leaf.

To delve deeper into the reversibility of the **ZT1** probe's response to Zn^{2+} and to assess its reversible imaging efficacy in Arabidopsis, we performed reversible imaging on the root tips and leaves under two distinct incubation conditions, Zn^{2+} solution (100 μM) and EDTA solution (100 μM).

As depicted in Fig. 6a, in the control group where only **ZT1** was applied, the fluorescence images exhibited faint fluorescence signals. However, upon the introduction of 100 μM Zn^{2+}, distinct fluorescence signals emerged in the root tips and leaves of Arabidopsis. Following the subsequent addition of 100 μM EDTA, the fluorescence signals vanished. The reintroduction of 100 μM Zn^{2+} then reinstated the fluorescence signals. These findings demonstrated that **ZT1** is capable of effectively detecting Zn^{2+} distribution in Arabidopsis root tips and leaves in real time and can accomplish reversible detection within the plant system. Besides, according to the statistical data (Fig. 6b and c), the same conclusion can be drawn.

Conclusion

In conclusion, we have developed a novel fluorescent probe, **ZT1**, based on 4-trifluoromethylcoumarin, which recognized Zn^{2+} in an "off-on" fluorescence mode through the photoinduced electron transfer (PET) mechanism. **ZT1** demonstrated high selectivity and water solubility for Zn^{2+} over other competing metal ions and exhibited high sensitivity, with a detection limit of 13.96 nM in HEPES buffer (pH =7.40). The complex formation, 1:1 stoichiometry, and binding mode were accurately determined using Job plot analysis, proton nuclear magnetic resonance (^1H NMR) titration, and electrospray ionization mass spectrometry (ESI-MS). **ZT1** has been successfully applied to the high-resolution visualization of Zn^{2+} in the root tips and leaf tissues of *Arabidopsis thaliana*, showcased excellent reversibility in recycling experiments. This probe offered a potent tool and novel insights for investigating Zn^{2+}-related physiological processes in plants, including their roles in growth, development, and metabolism, thus facilitated further in-depth exploration in this domain.

CRediT authorship contribution statement

Junyi Zheng: Methodology, Data curation. **Siyi Li:** Resources, Investigation. **Shunpeng Bai:** Software. **Xiao Liu:** Writing – original draft. **Ruitao Sun:** Project administration. **Shengzhen Xu:** Writing – review & editing. **Wei Qu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

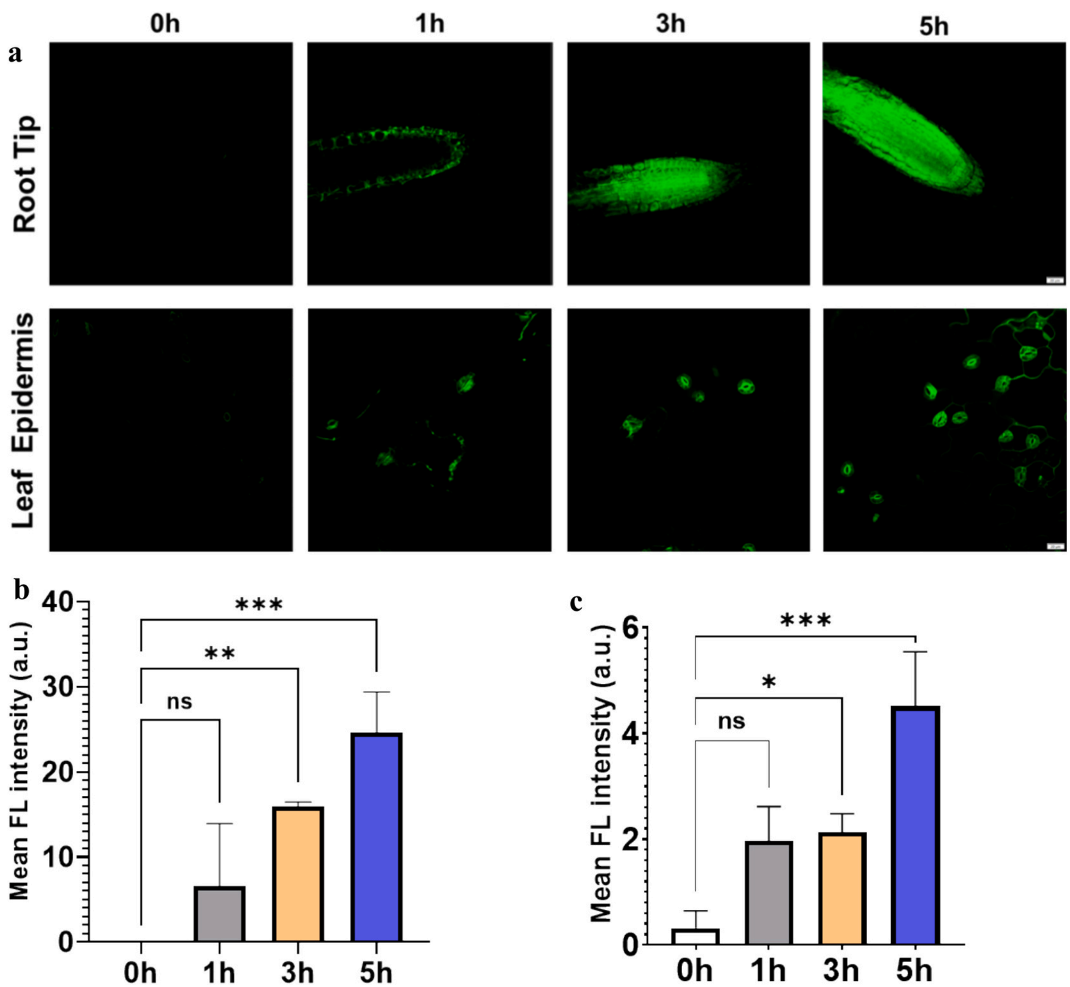

Fig. 5. a) Two-photon imaging of the probe (10 μM) in Arabidopsis root tips and leaves (Simple numbers $n = 3$, scale bar = 20 μm) at different time points; b) Average fluorescence intensity of Arabidopsis root tips; c) Average fluorescence intensity of Arabidopsis leaves.

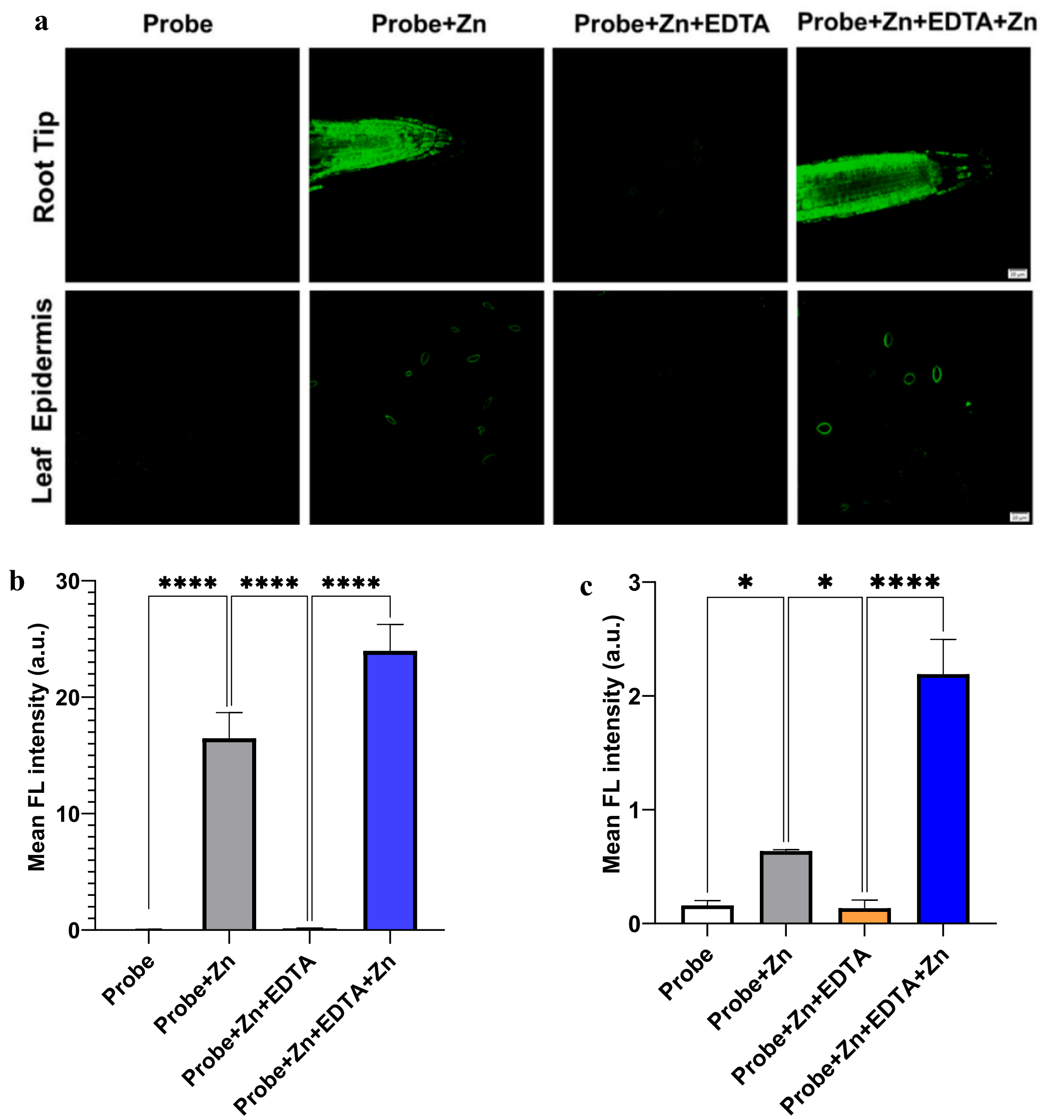

Fig. 6. a) Imaging of the probe (10 μM) in Arabidopsis root tips and leaves (Simple numbers $n = 3$, scale bar = 20 μm) in cyclic experiments; b) Average fluorescence intensity of root tips; c) Average fluorescence intensity of leaves.

the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tetlet.2025.155553.

Data availability

No data was used for the research described in the article.

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